

Generated Question Paper

Q1. A: Blood vessel C. [2 marks]

Q2. (d) Name two products of biotechnology, other than bread, that make use of microorganisms. [2 marks]

Q3. (ii) Name the process that causes gas bubbles to form in the dough. [1 marks]

Q4. Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]

Q5. The average photosynthesis rate of soybean plants was measured in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

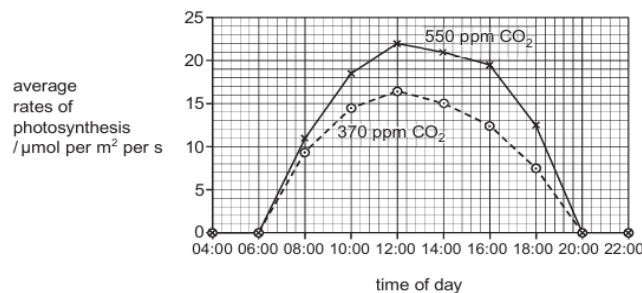


Fig. 2.1

Figure: photosynthesis_rate_graph.png

Q6. The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. [2 marks]

Q7. The nitrogen cycle is a process that occurs in the nitrogen cycle. [1 marks]

Q8. The name of organism A is a comma. [1 marks]

Biotechnology is used in the process of bread-making.

Fig. 1.1 shows some of the steps in making bread.

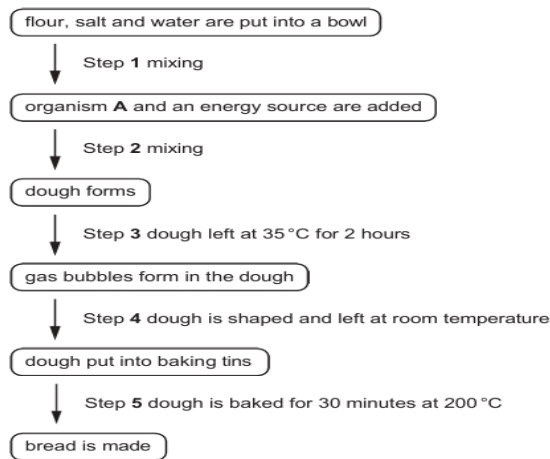


Fig. 1.1

Figure: bread_making_steps.png

Q9. (f) Discuss the effects of non-biodegradable plastics on terrestrial ecosystems. [5 marks]

Q10. (f) Describe the process by which a fertilised egg cell is fertilised. [3 marks]

Q11. (d) Explain why reflexes occur in unconscious people. [1 marks]

Q12. (iii) Name the gas that forms to create the gas bubbles in the dough. [1 marks]

Q13. The testes are a part of the human body. Meiosis is the process of dividing the testes into two parts. [3 marks]

One has been completed for you.

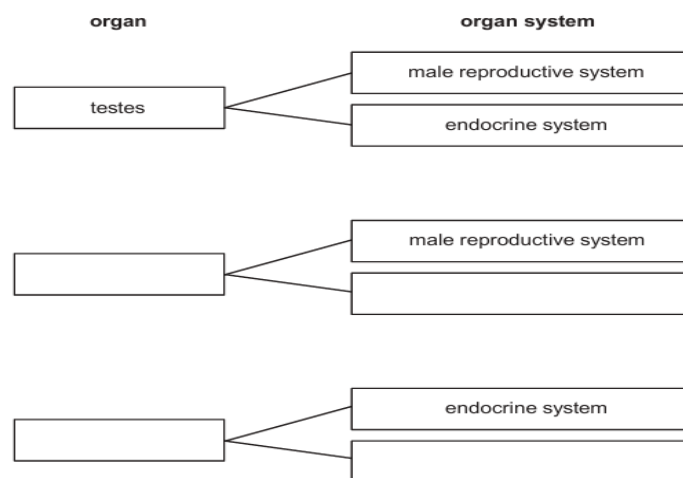


Fig. 5.1

Figure: organs_two_systems.png

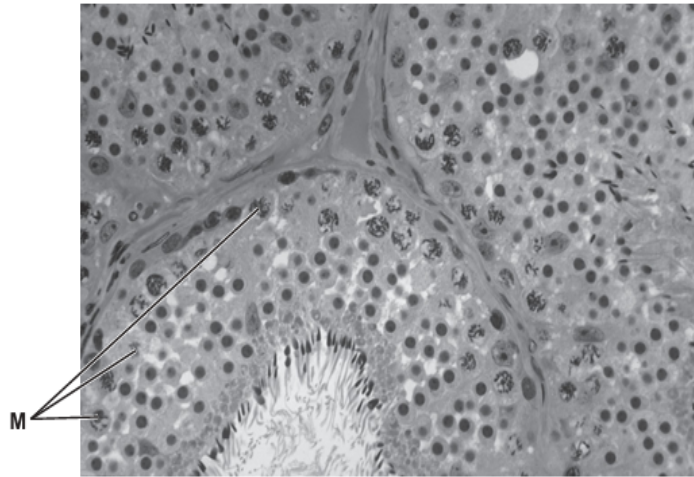


Fig. 5.2

Figure: mammalian_testis_micrograph.png

Q14. (e) shows a synapse and parts of two neurons. (e) begins center end center (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

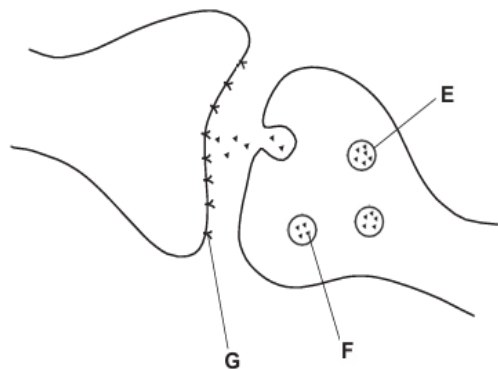


Fig. 4.1

Figure: synapse_diagram.png

Q15. The a) The possible causes of increased atmospheric carbon dioxide. [3 marks]

Q16. (ii) Enzymes are the chemical elements found in all proteins. [2 marks]

Q17. The retina is a part of the retina. What is the retina's function? [3 marks]

Q18. The effect of pH on lipase activity is shown on Fig.1. [6 marks]

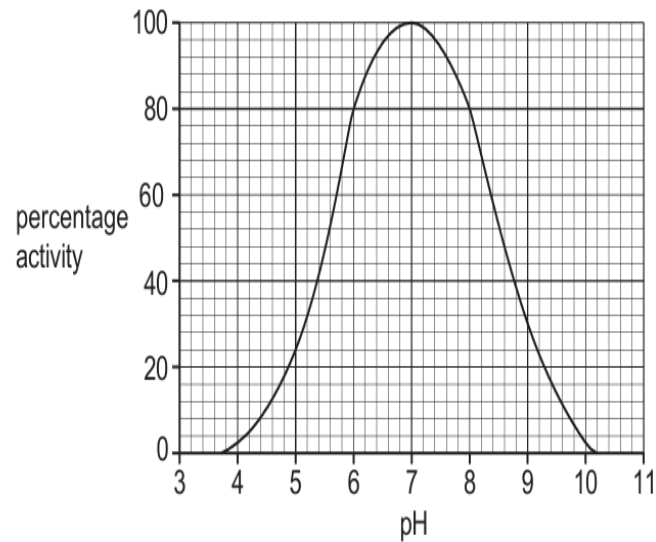


Figure: Fig1_1_LipaseActivity.png

Q19. (b) State the organs that make up the central nervous system. [1 marks]

Q20. The term 'species' is a term used to describe a species. [2 marks]



Fig. 6.1

Figure: potato_varieties.png

Q21. Fishing is a way to conserve fish stocks. [2 marks]

Q22. Plants growing in Europe begincenter begintabularcc toprule Plant Number & Leaf Area (cm\$2\$)
midrule 1 & 310 2 & 335 3 & 390 4 & 348 5 & 365 midrule Mean & 350 bottomrule endtabular endcenter (i)
Using the data in Table 5.2, calculate the mean leaf area for plants growing in North America. Give your
answer as a whole number and include the unit. [2 marks]

Q23. noindent_____ (ii) State a conclusion for the results shown in Table
..... [1 marks]

Q24. (a) State the hormone released from the testes. [1 marks]

Q25. (iii) Ancestors of monkey flower plants had a much smaller leaf area. Explain how the monkey flower plants have developed a larger leaf area over time. [5 marks]

Q26. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q27. (a) Describe how the structure of a neurone relates to its function. [3 marks]

Q28. The advantage of using a pyramid of energy is that it is more efficient. [2 marks]

Q29. Plants store starch in their leaves. [2 marks]

Q30. The temperature of the atoms in the nucleus of the atoms is the same as the temperature of the nucleus of the atom. [2 marks]

Q31. The right ventricle is located at the center of the heart. [1 marks]

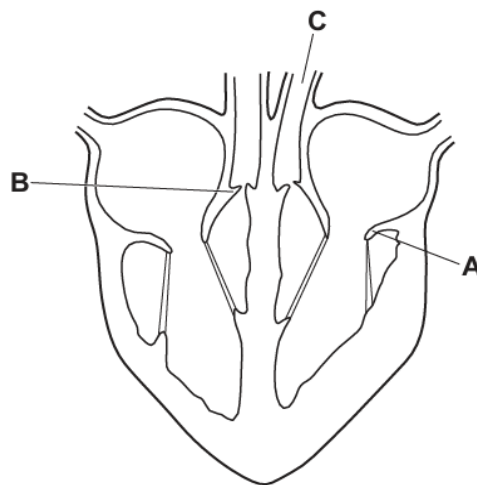


Fig. 2.1

Figure: Fig2_1_HeartCrossSection.png